Comparison of driving posture and position prediction methods for occupant packaging design – A case study

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Abstract: Accurately predicting driving postures and positions is crucial for occupant packaging design to ensure that vehicle interiors accommodate the intended and diverse driver populations. Digital human modelling tools typically play a critical role in occupant packaging design by providing functionality to predict the likely postures of different drivers in virtual environments. This paper presents a case study comparing two driving posture and position prediction methods: a statistical-based and an optimization-based method. The methods were compared against real-world data collected from two car models with different seat heights, involving 199 participants whose seat, steering wheel, and eye-point positions were measured. Patterns observed in the data, including method-specific deviations in fore-aft and vertical predictions, highlighted limitations in posture prediction accuracy and motivated this comparison. Application of these methods to real vehicles is rarely studied, and this case illustrates their performance in practice. The statistical method aligned better in the vehicle with a higher seat configuration, while the optimization method performed better in the lower seat configuration. Methodological insights based on the tested cases are outlined, providing guidance for refining posture prediction methods in DHM tools for occupant packaging design.

Keywords: occupant packaging design, digital human modelling, simulation, statistical regression, optimization

Introduction

Designing a vehicle interior to accommodate a wide range of drivers and passengers is the primary aim of occupant packaging design. Accurately predicting the seated posture and drivers' positions is crucial for effective occupant packaging design because it allows designers to optimize the vehicle interior for comfort, control accessibility, and visibility, among other ergonomic aspects, within the constraints of the vehicle geometry (Bubb et al., 2021). This predictive capability is particularly important as modern vehicles evolve to meet increasingly diverse user needs and expectations (Parida et al., 2020). Without accurate posture prediction, designers would have to rely on their own experience, previous designs, or extensive physical prototyping. These approaches are time-consuming, costly, and risk the accommodation of the target population (A. Wolf et al., 2020; Demirel et al., 2021; Perez Luque et al., 2022).

Integrating posture prediction capabilities into CAD and simulation tools in the design process provides designers with decision-support methods to develop vehicles that aim to meet ergonomic requirements. Further, to ensure inclusive, comfortable, and safe designs, human variability must be considered when predicting driver postures and key positions of their bodies, such as the H-point and eye-point. Variability can be related to differences in anthropometrics, behaviour, and individual preferences (P. Wolf et al., 2022; Jeong et al., 2023). For instance, shorter people typically sit more forward and higher within the seat adjustment range, while taller people usually prefer more rearward and lower positions within the seat adjustment range (Jonsson et al., 2008; Peng et al., 2018). Moreover, people with similar anthropometrics may behave and have different preferences, making them adopt substantially different driving postures (Peng et al., 2018). Consequently, driver prediction models should reflect these human differences in driving posture. Different driving posture models have been developed through data-driven and optimization approaches. Data-driven or *statistical* regression models generally predict the coordinates or body joint angles of specific positions of humans in the car interior. These regression models may consider different parameters from humans, such as sex, anthropometry, age, or body symmetry, and vehicles, such as vehicle class, seat designs, or driving venues (Reed et al., 2002; J. Park et al., 2016a, 2016b; Wu et al., 2022). In contrast, *optimization* methods predict human postures using optimization algorithms. It is often assumed that comfort or optimal joint angles are achieved when the torques across the joints are passively balanced. Consequently, postures can be predicted by assuming that individuals select postures that allow as many joints as possible to be close to the optimal angle (Yang et al., 2005; Hanson et al., 2006; Schmidt et al., 2014; S. J. Park et al., 2015). Thus, optimization works by minimizing deviations from the optimal angles within a search space usually restricted by defined constraints, such as roof height and available seat adjustment range, to achieve a desired prediction result. This process involves three essential components: determining the set of optimal joint angles, identifying the cost functions associated with deviations from the optimal posture, and determining how these costs are traded off or optimized (Reed et al., 2000).

Vehicle designers have traditionally used two- and three-dimensional accommodation tools provided by SAE International (formerly known as the Society of Automotive Engineers) for occupant packaging design. These tools help determine the layout of recommended seat positions, head contours, eye ellipses, and hand-reach envelopes to optimize driver ergonomics. However, due to the increasing complexity of occupant packaging design, designers have supplemented these traditional tools with three-dimensional human simulation tools over the past few decades, also known as digital human modelling (DHM) tools (Scataglini & Paul, 2019). DHM tools offer a broader framework for analyzing occupant packaging design, including tasks like ingress and egress, compared to SAE guidelines (Bubb et al., 2021). However, for posture prediction, SAE practices incorporate postural variability unrelated to driver or vehicle-specific variables, whereas the accuracy of DHM tools relies on the underlying assumptions of their predictive models, which can limit their accuracy and represent a general limitation in their application. An example of a DHM tool that has implemented a data-driven approach is Jack (Raschke & Cort, 2019), which uses a so-called cascade modelling approach, based on laboratory experiments with US subjects in specific vehicle package set-ups to predict driver positions (Reed et al., 2002). Another example of a data-driven approach implementation is a parametric CAD accommodation model for Fixed Heel Point (FHP) driving position, which provides geometric boundaries for equipped soldiers in military vehicles (Huston et al., 2024). In addition, Brolin et al. (2020) demonstrated that statistical prediction methods can be implemented in a DHM tool, which, at its core, has an optimization-based prediction approach. However, developing regression models that account for all human and vehicle parameters affecting seated driving posture is challenging for a data-driven approach due to the inherent complexity and numerous variables involved in driving situations, which must be accurately captured during data collection experiments. An example of DHM tools using an optimization approach is RAMSIS (Wirsching, 2019), with optimal angles based on a study by Seidl (1994). Limitations of the optimization approach include that the prediction models are not as accurate as the data-driven models for specific cases (Reed et al., 2000) and do not currently include anthropometry-specific comfort joint angles (Brolin et al., 2020).

While various posture prediction models have been proposed, few studies have explored how such models perform when applied in real vehicle configurations under industrial conditions. A notable exception is Reed et al. (2000), who compared prediction methods in a lab mock-up setting. However, with the growing integration of DHM tools in automotive workflows, it remains useful to examine how existing prediction approaches behave in specific contexts.

This paper presents a case study conducted in an automotive industrial setting to compare two existing prediction methods, a statistical regression-based method (SPM) and an optimization-based method (OPM), for seated driving posture. Rather than aiming to generalize to all vehicle types or populations, the objective is to examine how each method performs in relation to real-world data from two vehicles with distinct seat configurations. Through this comparison, the case study aims to provide insights into the strengths and limitations of each approach when applied in a specific design scenario. This case-based approach contributes to understanding how DHM prediction methods function in practice and supports reflection on their selection and application in vehicle interior design.

Method

Data collection context

This case study was conducted in collaboration with an automotive manufacturer, using real-world data collected in two Volvo vehicle models. The vehicles represent contrasting seating configurations relevant for occupant packaging design considerations: Vehicle A featured a lower seat configuration, H30=265 mm and L6=563 mm, and Vehicle B a higher seat configuration, H30=310 mm and L6=540 mm (all vehicle dimension definitions according to SAE J1100, 2009) (Figure 1). The test participants were asked to adjust the seat and steering wheel positions to find their preferred driving position before and during a driving session of 10 minutes in a real traffic situation. This was repeated three times, and the seat position and steering wheel position data were collected after the third driving session for each test participant. Before each trial, the seat was returned to an extreme or "not usable" position to trigger the participant to re-adjust.

Figure 1. Volvo car models used in the user study: Vehicle A (left) and Vehicle B (right).

The data collection followed an established industrial procedure, used internally at the company to evaluate occupant packaging layouts and to examine how posture prediction methods that are typically developed and evaluated in lab environments perform under actual usage conditions. The procedure for collecting H-point, eye-point, and steering wheel point data builds upon the approach described by Lejon & Thorsén (2013). Three-seat adjustment motors were employed to determine each test participant's H-point based on the seat position and adjustment settings. The seat tolerances specific to each car model were also considered when determining the H-point. The mid-eye position was captured with a photo from a fixed side-view camera in the car while participants were driving straight forward during the third driving session. Lastly, the steering wheel position was manually measured in the cars for all test participants.

Test participants were recruited from Volvo Cars staff, and they were not involved in the study's planning, design, or execution. In total, 104 participants were involved in collecting data for Vehicle A (study conducted in 2017), and 95 participants were involved in collecting data for Vehicle B (study conducted in 2018). The anthropometric data of the participants is described in Table 1.

Table 1. Summary of anthropometric data of participants involved in the case study.

	Vehicle A				Vehicle B			
Gender	**Females**		**Males**		**Females**		**Males**	
Subjects	44		60		39		56	
Anthropometric measurements	**Mean**	**SD**	**Min**	**Max**	**Mean**	**SD**	**Min**	**Max**
1 Stature without shoes	1730	128	1460	1981	1735	150	1460	2030
2 Weight (kg)	72	17	44	148.5	74	20	44	138.5
3 Waist circumference, standing	857	110	665	1260	868	127	630	1270
4 Hip width, standing	351	23	300	424	353	27	300	422
5 Upper arm length	353	31	300	428	355	39	295	454
6 Elbow-fingertip	455	40	374	550	456	45	365	560
7 Sitting height	920	58	802	1045	920	66	802	1106
8 Hip width, sitting	383	26	334	478	385	31	335	481
9 Buttock to front of knee	601	51	496	756	606	59	496	756
10 Buttock to back of knee	486	43	407	615	489	50	405	615
11 Knee height sitting, without shoes	531	53	439	667	538	60	440	699
12 Foot width without shoes (left)	94	9	74	113	95	9	74	114
13 Foot length without shoes (left)	249	22	206	304	250	26	189	312

Note. In millimetres (mm) unless otherwise noted.

Participants were selected to reflect a broad range of body measurements rather than to represent a population sample, with emphasis on including both shorter and taller individuals. As the collective anthropometric diversity of the participants did not follow a normal distribution, they were divided into three groups based on height: Short (1460–1634 mm), Medium (1635–1795 mm), and Tall (1796–2030 mm). Each group comprised approximately 1/3 of the total number of participants, ensuring a balanced representation of different statures. No separation was made between male and female participants regarding height distribution, as the analysis focused on stature regardless of gender.

Statistical Prediction Method (SPM)

The driver posturing process involves several parameters where both the steering wheel and seat can move independently of each other. In which order this process is done might vary between vehicles and drivers, i.e. it is not certain that the steering wheel position is set before the seat is positioned or vice versa. The positioning of the steering wheel and seat is rather done iteratively, more or less in relation to each other. In the study of this paper, the intention is to predict the position of all adjustable parts of the driver interior within given constraints, such as adjustment ranges of the steering wheel and seat. To implement a statistical prediction method that includes both prediction of steering wheel position and seat position, a specific sequence of these predictions needs to be decided. The statistical prediction method (SPM) uses two different prediction models in a sequence of three steps: 1) first, a prediction of the fore-aft steering wheel position is done (Reed, 2013), 2) followed by a prediction of the H-point position (Park et al., 2016b), 3) to eventually be able to predict the position for the centre-eye point (Park et al., 2016b). The fore-aft steering wheel position (L6) is a dimension that is set for each vehicle (SAE J3099, 2024). The steering wheel can usually move within an adjustment range, where L6 is defined at the centre of this range. The prediction of the steering wheel location is made by using a model that was initially developed as an ordinal logistic regression to predict the distribution of subjective responses as a function of fore-aft steering wheel position (L6), seat height (H30), and driver stature (Reed, 2013). In the study, subjective responses were measured on a rating scale that varied from 1 to 7. Response 4 was counted as "just right" and in this study inserted into a transformed linear part of the logistic model to be able to predict a preferred fore-aft steering wheel location (L6) (mm) as

$$L6 = \frac{4+0.003441\times \text{Stature}+0.01854\times \text{H30}-0.00004958\times H30^2}{0.021182} \tag{1}$$

If the predicted preferred fore-aft steering wheel location (L6) is outside the adjustment range, then the location is moved to the closest point of the adjustment range area. The predicted preferred fore-aft steering wheel location is subsequently used as the L6 value, instead of the actual L6 value of the vehicle, to predict coordinates for the preferred seat position, i.e. the H-point, with an x-coordinate in relation to the Ball of Foot point (BOF) and a z-coordinate in relation to the ankle heel point (AHP). In the study by Park et al. (2016b), the L6 value was set in different predefined positions, and participants were not allowed to change it. As a result, the prediction models could use different L6 values as predictive variables, where the recommendation is to use the vehicle L6 value in the centre of the steering wheel adjustment range. The motivation for using a predicted steering wheel location instead of the actual vehicle L6 is to statistically determine the L6 position when no predefined value is available, as well as to capture the variability found in real data, thereby reflecting a wider range of outcomes in subsequent predictions. In the third step, coordinates for the centre-eye position are predicted in relation to the H-point position. Following the method by Park et al. (2016b), different posture prediction models for men and women were used (Table 2).

Table 2. Driver Posture-Prediction Models for the Cascade Modelling Approach (Park et al., 2016b).

	Female driver posture-prediction	
Dependent Variable	**Regression Model (mm)**	**RMSE (mm)**
H-point x re PRP	$678 + 0.284 \times \mathrm{S} - 494 \times \mathrm{SHS} + 2.33 \times \mathrm{BMI} - 0.388 \times \mathrm{H30} + 0.426 \times (\mathrm{L6} - 600)$	24.9
H-point z re AHP	$129 - 6.2 \times 10^{-2} \times \mathrm{S} + 0.909 \times \mathrm{H30} - 4.65 \times 10^{-2} \times (\mathrm{L6} - 600)$	12.4
Centre-eye x re H-point	$-364 + 7.64 \times 10^{-2} \times \mathrm{S} + 540 \times \mathrm{SHS} + 0.124 \times (\mathrm{L6} - 600)$	37.7
Centre-eye z re H-point	$-461 + 0.163 \times \mathrm{S} + 1445 \times \mathrm{SHS} - 15.1 \times \mathrm{BMI} + 3.61 \times \mathrm{Age} + 9.93 \times 10^{-3} \times \mathrm{S} \times \mathrm{BMI} - 6.54 \times \mathrm{SHS} \times \mathrm{Age}$	12.7
	Male driver posture-prediction	
Dependent Variable	**Regression Model (mm)**	**RMSE (mm)**
H-point x re PRP	$-48 + 0.56 \times S + 1.39 \times BMI + 7.53 \times Age - 0.42 \times H + 0.505 \times (\mathrm{L6} - 600) - 3.98 \times 10^{-3} \times S \times Age$	25.3
H-point z re AHP	$123.1 - 5.864 \times 10^{-2} \times S + 0.945 \times H30$	10.6
Centre-eye x re H-point	$-578 + 0.174 \times \mathrm{S} + 706 \times \mathrm{SHS} - 1.55 \times \mathrm{BMI} - 6.48 \times 10^{-2} \times \mathrm{H30}$	32.1
Centre-eye z re H-point	$-498 + 0.361 \times \mathrm{Stature} + 845 \times \mathrm{SHS} + 2.76 \times \mathrm{BMI} - 0.175 \times \mathrm{Age}$	14.4

Note. S = stature (mm); SHS = sitting height/stature; BMI = body mass index (kg/m^2); H30 = seat height (mm).

Optimization Prediction Method (OPM)

The second method for predicting driving postures involves an optimization approach. In this study, a research version of the DHM tool IPS IMMA was used to represent the optimization prediction method for occupant packaging design (Hanson et al., 2019). The optimization model employed in the software is based on the principles described by Bohlin et al. (2012) and Delfs et al. (2013). This model predicts the driver's posture by solving an optimization problem that seeks to maximize a comfort function, which evaluates ergonomic factors such as joint angles, joint torques, and proximity to obstacles. The optimization is expressed as:

$$\text{maximize } h(x) \qquad \text{while: } g(x) = 0,$$

where h(x) represents the comfort function, prioritizing postures that minimize joint strain and maintain ergonomic alignment, and g(x) refers to kinematic constraints such as body segment alignments (positioning constraints) and biomechanical balance. The driver's optimal joint angles, which are the basis of the optimization in the software, were originally derived from a study by Bergman et al. (2015) and adjusted to fit the biomechanical model of IPS IMMA. In addition, collision avoidance is also included by penalizing postures where body parts approach environmental obstacles. The iterative optimization process refines the posture prediction until it satisfies all the constraints while maximizing the comfort function.

The DHM scene consisted of a group of manikins and the geometries of each of the two vehicle types. The manikin family corresponding to each vehicle represents drivers who participated in the data collection, i.e. manikins were created with the same anthropometric measurements as the participants (Table 1). Then, a standard DHM simulation setup was followed to make manikins adopt an initial seated driving posture (Perez Luque et al., 2022). Several basic constraints were established to ensure that the manikins could perform the necessary driving tasks. These constraints included positioning the feet on the pedals, placing the hands on the steering wheel, ensuring that the hip joint falls within the H-point envelope, and aligning the torso with the mid-back seat using attachment points in the DHM tool (Figure 2). These attachment points restrict the manikins to only move within a defined two or three-dimensional range (such as a line, a plane, or a box), as illustrated in Figure 2. Furthermore, four additional constraints were implemented to improve the accuracy of posture predictions specific to vehicle environments using the DHM tool. These constraints were necessary to address limitations in the DHM tool's default predictive capabilities and better reflect the ergonomic requirements of automotive settings. Three of these constraints addressed collision avoidance with specific vehicle geometries. They were handled using the collision avoidance functionality of the software and included the head-to-roof distance, knee-to-vehicle geometry distance, and thigh-to-steering wheel distance (Figure 2). The fourth constraint, which ensures a minimum downward vision angle, was implemented in the software with the attachment point functionality (Hanson et al., 2019) (Figure 2). This attachment inclination followed the slope of the hood. Then, the "Driving Car" strategy in the DHM tool was selected to make the manikins adopt the driving posture prediction following the OPM.

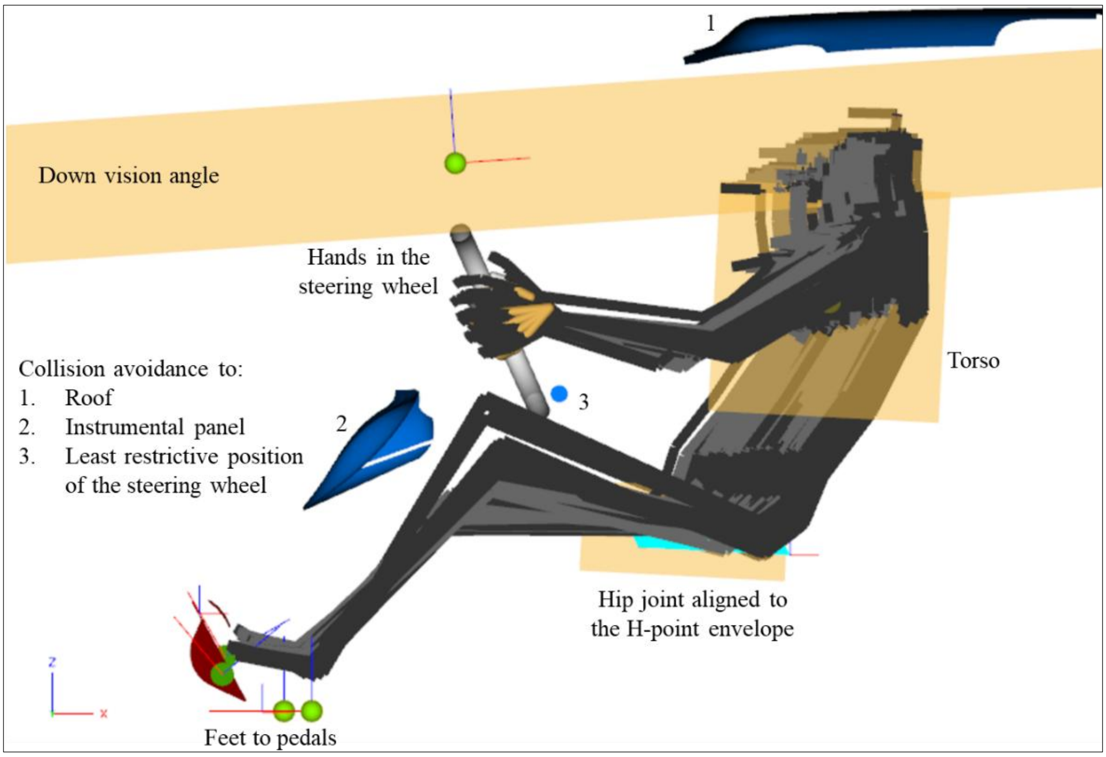


Figure 2. DHM tool scene setup for posture prediction in the case study: manikins seated with basic and additional constraints. Basic constraints ensured necessary driving tasks functionalities (e.g., feet on pedals, hands on wheel), while additional automotive-specific constraints improved realism (e.g., collision avoidance and vision angle).

Results

This section outlines the driving positions obtained for each manikin group in vehicles A and B using the two prediction methods, compared to the real-world collected data. The results show the location of driver key variables: the H-point, eye-point, and steering wheel position.

H-point prediction

Figure 3 illustrates the predicted H-point using SPM (blue points) and OPM (orange points) against the real collected data (green points) for Vehicle A and Vehicle B. Figure 3 also shows results when residual variance (RV), i.e., a degree of randomness, is included in the SPM using the RMSE values given by Park et al. (2016b) together with a stochastic component from a standard normal distribution. Additionally, Figure 3 also illustrates results for the OPM when loosening constraints (LC) are applied, specifically a doubled range of the hip joint attachment in the Z direction.

Vehicle A

In the x-axis (fore-aft direction), there is a noticeable trend where OPM predicts the H-point further back, while SPM predictions tend to be more forward (Figure 3). In this way, OPM predictions could be unable to accurately predict more forward positions, whereas SPM could be failing to cover some of the people who sit further back within the H-point envelope. This tendency of forward prediction by SPM and rearward prediction by OPM is consistent across most data points for Vehicle A, indicating a potential tendency of these methods to position the driver slightly further forward (SPM) or back (OPM) than the real-world measurements. In the z-axis (vertical direction), OPM tends to predict a higher H-point than SPM (Figure 3). SPM's z-axis predictions exhibit a narrower spread in the z-direction, whereas OPM captures a wider range of z-axis positions (Figure 3). However, this increased variability does not necessarily correlate with the actual measurements, indicating a potential overestimation in vertical positioning. This suggests that both methods struggle to accurately capture the variability in vertical seating positions, with OPM overestimating the vertical spread and SPM not fully representing the vertical spread.

Table 3 further details these observations with statistical measures. The average error for the x-axis prediction shows that SPM has a negative bias (-28.4 mm), indicating it generally predicts positions slightly forward compared to the real data. OPM, on the other hand, has a positive bias (27.2 mm), confirming its tendency to predict positions further back. The correlation coefficients for both methods are high (0.8 and 0.9), indicating a strong linear relationship with the actual data. However, the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) show that OPM has slightly better accuracy in the x-axis predictions. For the z-axis, SPM exhibits a lower average error (-2.1 mm), indicating it predicts better the vertical H-point position. OPM shows a slight positive bias (6.7 mm), suggesting that it predicts the H-point slightly higher on average than it actually is. The correlation for SPM is higher (0.7) than for OPM (0.4), indicating better predictive reliability in the vertical dimension. This is verified with both MAE and RMSE values, showing more pronounced errors in OPM predictions.

Vehicle B

For Vehicle B, similar patterns are observed as in Vehicle A, with SPM predicting the H-point further forward and OPM predicting it further back and higher in the z-axis (Figure 3). However, in Vehicle B, the errors are larger for both methods and directions, particularly for SPM in the x-axis predictions, showing a more pronounced forward prediction (-50.3) (Figure 3).

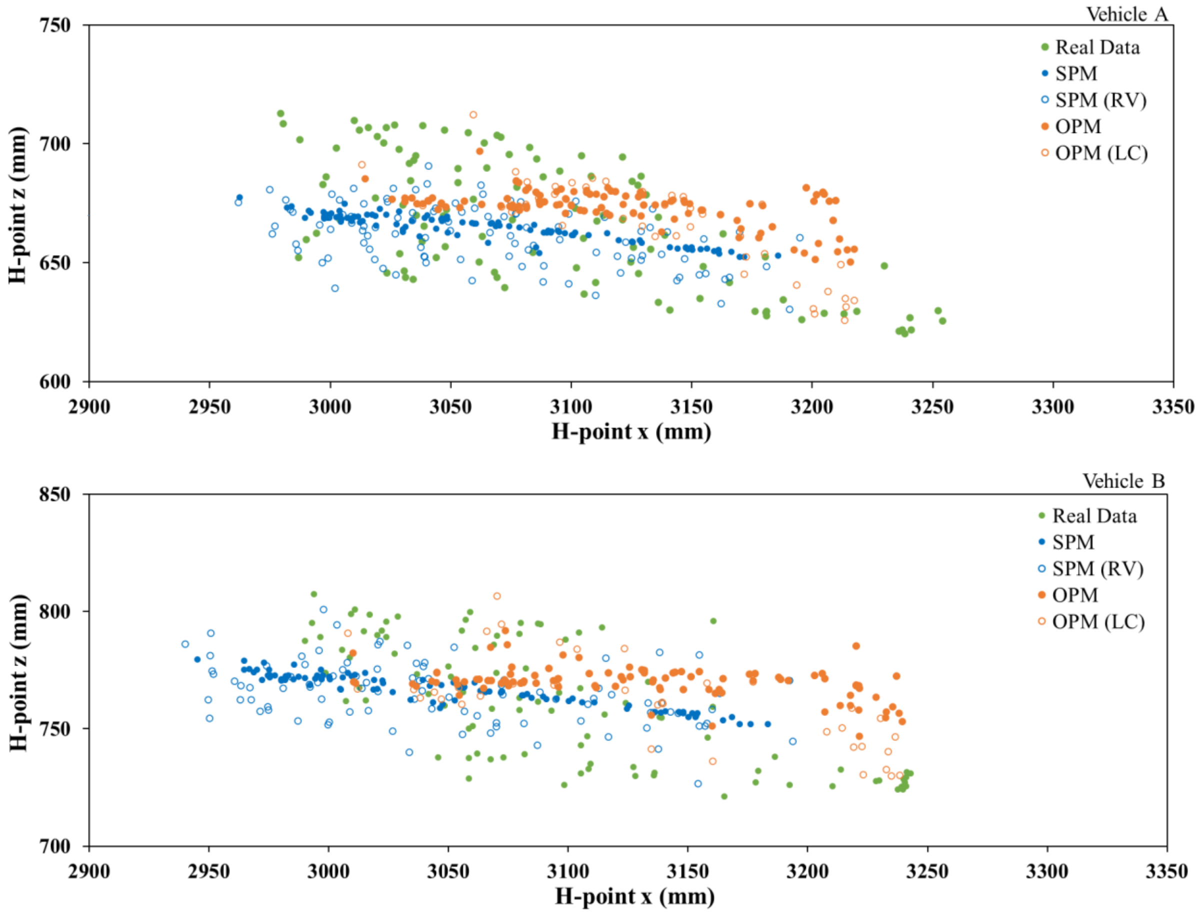


Figure 3. Comparison of H-point prediction by SPM (blue points) and OPM (orange points) against real collected data (green colour) for Vehicle A (top) and Vehicle B (bottom).

Table 3. Comparison of prediction methods for H-point: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		**Correlation**		**MAE**		**RMSE**	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
H-point x (mm)	SPM	-28.4 (40)	-50.3 (33)	0.8	0.9	39.7	53	48.8	59.9
	OPM	27.2 (75)	31.9 (37)	0.9	0.9	38.3	38.4	47.2	48.5
	SPM (RV)	-27.2 (45)	-51.5 (38)	0.8	0.9	42.3	55.8	52.3	63.7
	OPM (LC)	27.2 (39)	31.9 (37)	0.9	0.9	38.3	38.4	47.2	48.5
H-point z (mm)	SPM	-2.1 (23)	6.3 (21)	0.7	0.8	20.4	19.7	23.4	22.3
	OPM	6.7 (26)	9.4 (25)	0.4	0.4	22.6	22.6	26	26.2
	SPM (RV)	-5.1 (24)	4.9 (24)	0.4	0.4	20.9	20.5	24.8	24.5
	OPM (LC)	4.7 (24)	7.1 (24)	0.4	0.4	20.8	21.1	24.8	25.4

Eye-point prediction

Figure 4 and Table 4 compare the predicted eye-point positions using SPM and OPM against the collected data for Vehicles A and B, respectively.

Vehicle A

SPM predicts a more forward eye-point position compared to the real data, with an average error of -60.3 mm, while on the other hand, OPM predicts a more rearward position with a lower average error of 13 mm (Figure 4, Table 4). However, neither method fully captures the spread of the real data in the x-axis, indicating limitations in both predictive models. The correlation for both methods in the x-axis is 0.7, showing a moderate linear relationship with the actual data. In addition, OPM shows lower error values in predicting the eye point in the x-axis than SPM.

In contrast, the z-axis predictions are relatively similar in performance, although OPM shows slightly lower error values, suggesting a marginally better fit to the real data.

Vehicle B

SPM predicts a more forward position, with an average error of -62.7 mm, while OPM predicts a much more rearward position with a significant error of 107.4 mm, indicating a poor fit for Vehicle B (Figure 4, Table 4).

On the z-axis, SPM achieves a low average error of -1.9 mm, closely matching the real data, while OPM shows a slight overestimation at 6.5 mm. The correlation for SPM and OPM on the z-axis is equal (0.8). However, SPM reflects better predictive accuracy in the vertical dimension observing lower error values.

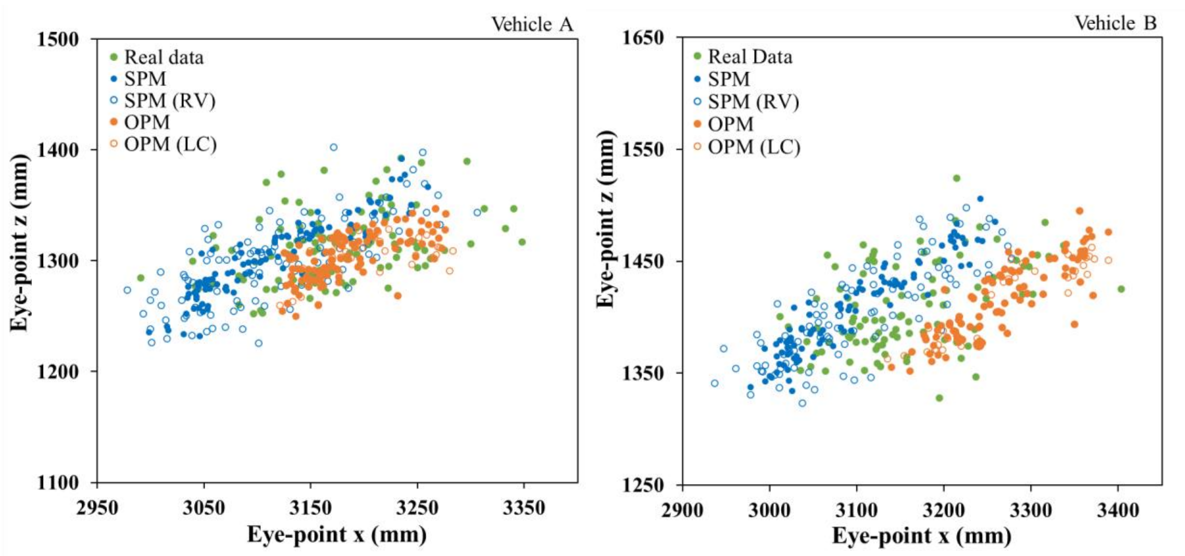


Figure 4. Comparison of eye-point predictions by SPM (blue points) and OPM (orange points) against real collected data (green points) for Vehicle A (left) and Vehicle B (right).

Table 4. Comparison of prediction methods for Eye-point: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		Correlation		MAE		RMSE	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
Eye-point x (mm)	SPM	-60.3 (52)	-62.7 (63)	0.7	0.7	67.8	74.4	79.1	88.9
	OPM	13.0 (72)	107.4 (60)	0.7	0.7	39.9	110	52.3	122.8
	SPM (RV)	-57.4 (61)	-56.2 (74)	0.6	0.6	69	74.4	83.4	92.5
	OPM (LC)	14.3 (51)	107.8 (60)	0.7	0.7	40.2	110.3	52.3	123.2
Eye-point z (mm)	SPM	-10.7 (28)	-1.9 (24)	0.7	0.8	24.5	18.3	29.7	23.8
	OPM	-10.0 (38)	6.5 (26)	0.6	0.8	22.5	19.7	28.5	26.4
	SPM (RV)	-14.3 (32)	-4.6 (27)	0.6	0.8	28.5	21.8	34.8	27.7
	OPM (LC)	-12.0 (29)	4.4 (27)	0.5	0.7	24	19.9	31.2	27.3

Steering wheel position prediction

Table 5 shows the steering wheel position predictions in the x-axis for both Vehicle A and Vehicle B using SPM and OPM, respectively.

Vehicle A

For Vehicle A, OPM closely aligns with the real data, showing an average error of 4.7 mm, while SPM has a more pronounced forward bias with an average error of -18.1 mm (Table 5). OPM demonstrates slightly better predictive accuracy, with a higher correlation (0.6) compared to SPM (0.5) and lower error values for both MAE (15.2 mm) and RMSE (18.1 mm), indicating a better overall fit.

Vehicle B

For Vehicle B, OPM slightly outperforms SPM, with an average error of 8.3 mm compared to SPM's -10.2 mm (Table 5). The correlation for both methods is lower in Vehicle B, with OPM achieving 0.4 and SPM only 0.1, reflecting greater variability in predictive accuracy. OPM also shows lower MAE (18.4 mm) and RMSE (22 mm) than SPM.

Table 5. Comparison of prediction methods for Steering wheel position: Average Error (mean of the predicted minus measured) and Standard deviation (SD) of the average errors; Correlation; Mean Absolute Error (MAE); and Root Mean Squared Error (RMSE) between observed and predicted data.

		Average Error (SD)		**Correlation**		**MAE**		**RMSE**	
		Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B	Vehicle A	Vehicle B
Steering wheel position x (mm)	SPM	-18.1 (18)	-10.2 (26)	0.5	0.1	20.5	22	25.8	27.8
	OPM	4.7 (18)	8.3 (20)	0.6	0.4	15.2	18.4	18.1	22

Results summary

The comparison of SPM and OPM revealed that each method demonstrates strengths depending on the specific vehicle configuration and prediction point. Boxplots of the Euclidean distances for SPM and OPM predictions are also included in the Appendix, offering a complementary visualization of the methods' overall prediction accuracy.

Table 6 summarizes the best-performing method in this case study based on all evaluation metrics across both vehicles, highlighting that each method shows better alignment depending on the specific vehicle configuration. While Table 6 presents an overview of the methods' performance, additional insights are available through Bland-Altman plots, which are included in the Appendix providing a more detailed analysis. These plots visually represent the differences between predicted and measured values, helping to identify key trade-offs between precision and adaptability for each method. Boxplots of the Euclidean distances for SPM and OPM predictions are also included in the Appendix, offering a complementary visualization of the methods' overall prediction accuracy.

Table 6. Best performing prediction method for Vehicle A and Vehicle B.

	Best prediction method	
	Vehicle A	**Vehicle B**
H-point x	OPM	OPM
H-point z	SPM	SPM
Eye-point x	OPM	SPM
Eye-point z	OPM	SPM
Steering wheel x	OPM	OPM

Discussion

This case study compares two different methods, SPM and OPM, for predicting seated driving postures and positions within vehicle occupant packaging design. SPM relies on human anthropometrics to predict the location of driver key variables such as the H-point, eye-point, and steering wheel position through a cascade modelling approach, where predictions are made sequentially. However, its base on fixed equations limits adaptability to additional constraints or variations outside the predefined parameters. In contrast, OPM assumes individuals select postures that align their joints as closely as possible to optimal joint angles, maximizing their comfort. This approach is able to adapt to the constraints defined within the DHM scene. However, OPM prediction models may require refinement to represent postural variability across targeted populations better.

Results

The comparison between SPM and OPM for Vehicles A and B reveals differences in how each method performs across the key predicted points: H-point, eye-point, and steering wheel position.

H-point prediction

For the x-coordinate of the H-point, OPM shows better accuracy in the tested scenarios for both vehicles, outperforming SPM. Within the scope of this case, this suggests that OPM is more effective in predicting the fore-aft positioning of the occupant's hip joint for both the low (Vehicle A) and high (Vehicle B) seat configurations. Additionally, the Bland-Altman plots reveal additional insights. For Vehicle A, SPM demonstrates narrower limits of agreement (LOA), indicating better precision, despite having a slight underprediction bias. In contrast, OPM shows wider LOA and more random scatter, which reflects greater adaptability to diverse driver postures but lower precision. A similar trend is observed in Vehicle B, where SPM maintains higher precision but exhibits a stronger underprediction bias, while OPM performs with less systematic bias but greater variability. When considering the z-coordinate (vertical positioning), SPM performs better across both vehicles, which corresponds with the Bland-Altman analysis. This variation in performance between axes highlights the need for further development or complementary consideration of the two methods for H-point predictions.

To further investigate the poor performance of OPM in the z-axis for H-point prediction, the population was divided into three stature groups: short, medium, and tall. This was conducted to determine whether errors increase for specific stature groups, given that stature is a key aspect influencing where individuals sit within the H-point envelope (P. Wolf et al., 2022). Figure 5 shows that the difference error between the measured and predicted increases significantly for OPM as stature increases across both Vehicle A and Vehicle B. While OPM can more accurately predict H-point vertical positions for average-sized occupants, its predictive capability diminishes for individuals at the extremes of the stature range, especially taller drivers. SPM displays similar trends with less pronounced errors, resulting in a more consistent error distribution across all stature groups, making it a more reliable method for predicting H-point z-coordinates in vehicles where accommodating a wide range of driver statures is essential.

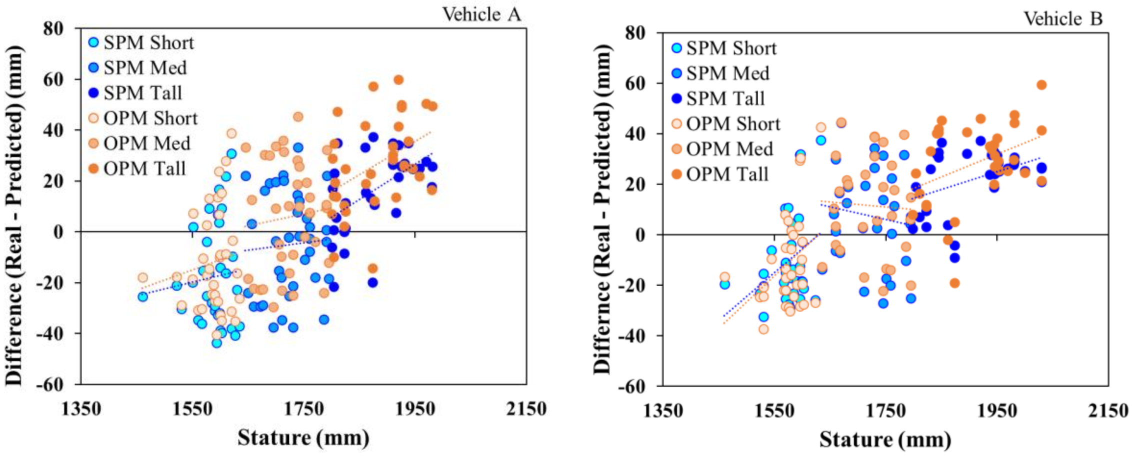


Figure 5. H-point z: Difference errors trend line across stature.

Eye-point prediction

The eye-point prediction results follow another pattern. For Vehicle A, OPM outperforms SPM in both the x- and z-axes. This suggests that OPM is more reliable in predicting eye-point locations for specific vehicle configurations similar to Vehicle A. For Vehicle B, which has a different vehicle configuration with a higher seat height, SPM performs better in both axes. This shift in prediction accuracy suggests that SPM potentially captures better the driver's seating posture and eye-point positioning in vehicle configurations similar to Vehicle B. However, the Bland-Altman plots provide additional context, showing that while OPM performs better according to metrics for Vehicle A, SPM demonstrates narrower LOA and greater precision for both x- and z-coordinates. For Vehicle B, SPM outperforms OPM across all analyses, including Bland-Altman.

Steering wheel position prediction

For the steering wheel x-coordinate, OPM again provides higher accuracy in Vehicle A and Vehicle B, indicating its potential ability to adapt to posture and steering wheel positioning requirements of vehicles with lower and higher seat heights, respectively.

Prediction method by vehicle type

Overall, considering the evaluation metrics, OPM showed better alignment with measured data for Vehicle A, which features a lower seat height in the tested scenarios. Its performance in predicting the H-point, eye-point, and steering wheel position in the x-axis suggests that the current underlying set of optimal angles used in OPM suits better the specific ergonomic configuration of lower-seated vehicles like Vehicle A. Nonetheless, the Bland-Altman analysis highlights that SPM's narrower LOA could result in better precision, even in cases where systematic bias is greater. For Vehicle B, which has a higher seat height, SPM outperforms OPM across most prediction points, particularly in the vertical (z-axis) predictions, although it does not necessarily predict the variability in it. This could indicate that the default set of optimal angles in OPM does not account for the variations in drivers' posture resulting from different vehicle configurations like Vehicle B. Consequently, a different set of optimal angles tailored for higher seat configuration vehicles could improve OPM's prediction performance. Such adjustment could potentially bring OPM's results in Vehicle B closer to the accuracy it achieves in Vehicle A with a lower seat position, suggesting its potential prediction utility across different vehicle types.

Limitations

While the study provides valuable insights, several limitations must be acknowledged. These relate to the number of vehicles analyzed, data collection protocols, and the methods and tools evaluated. First, the study was conducted using two vehicle configurations, which represent contrasting but limited configurations. Given that this is a case study, the results should be interpreted within these specific contexts, and the generalizability of the findings to other vehicle types is limited. Future studies should incorporate a broader range of vehicle configurations to evaluate the methods more comprehensively and enable the generalizability of the results. Additionally, differences in how companies collect or estimate real-world posture data, such as alternatives in pose estimation methods or motion capture technologies, may influence the observed outcomes and their alignment with predictive models. Such methodological variation could result in different patterns and should be taken into account when comparing across industrial contexts.

Second, potential biases may have arisen from the initial seat placement during data collection, as this could have influenced how participants adjusted and selected their seating positions. Although a neutral starting position was adopted based on standard methods from previous research, a recent study has shown that starting conditions can introduce biases (Jung & Park, 2023). However, it is important to note that the methodology for collecting posture data was based on an established methodology used by the company conducting the data collection, where consistency and comparability of results are prioritized, and direct measurement in real vehicles is a key component. Future studies should consider methodologies that reduce these effects to address this, such as conducting multiple trials with varied initial seat positions.

Third, this study was limited to evaluating OPM implemented in a single DHM tool, IPS IMMA. Expanding the analysis to include OPM implemented in other DHM tools could provide a broader understanding of the method's effectiveness and applicability across different DHM tools. Additionally, limitations in the specific IPS IMMA version used, such as the body models and their meshes, could have affected the accuracy of the predictions as they may not realistically represent humans. Enhancing the fidelity of these body models could lead to more accurate and reliable results, particularly in capturing the detailed postural variations of different body sizes.

Prediction methods improvements

While both methods show potential for predicting driver posture in occupant packaging design, their performance varies depending on the specific vehicle configuration. Both methods leave room for improvement, emphasizing the importance of further refinement and validation of predictive models to improve their reliability and applicability. On the one hand, an advantage of SPM is that the prediction of characteristic driving positions is written as closed-form, linear equations (Table 2), which makes them independent of any human model and easier to use. In contrast, OPM must be implemented using a particular kinematic linkage or digital human model. On the other hand, one potential advantage of OPM is that it may provide more generalizable results to new design tasks as its results are a consequence of the digital human model and constraints defined in the virtual environment, whereas SPM is not able to adapt to additional constraints or aspects to the ones considered in their equations. Furthermore, analyzing a specific vehicle configuration can be crucial when choosing prediction methods. This study shows that the set of optimal joint angles for OPM used in this study may yield better prediction results for vehicles with lower seat configurations, while SPM shows better performance for vehicles with higher seat configurations.

While both SPM and OPM present distinct advantages depending on the application, there remain areas where each method could be refined to better capture the complexity of human posture across different vehicle configurations. SPM could be enhanced by adjusting the results using the average error found compared to real world data. Based on the average error of this initial evaluation, adjustments could be made following the same sequence as the cascade modelling approach. Thus, the average error for the steering wheel position would be calculated first, and the adjustment of those predictions would be added before the average error of the H-point position could be calculated next. Lastly, the average error of the centre-eye position could be calculated after adjustments to predictions of the H-point have been added. Different adjustments could be calculated for different vehicle models.

As observed in Figure 5, OPM's error performance particularly increases depending on the stature, suggesting that the optimal angles used in the optimization process could be further refined for specific manikin statures and vehicle configurations. As it currently works, all manikins in a family follow the same optimization values without considering different postures for different body dimensions and postural variability, as unlikely happens in reality (Kyung & Nussbaum, 2009; J. Park et al., 2016b; Brolin et al., 2020; Bubb et al., 2021). Developing a more tailored set of optimal driving angles that consider the unique body dimensions and postural tendencies of different stature groups and for specific types of vehicles could enhance the accuracy of OPM. For example, developing separate sets of optimal angles for distinct manikin sizes or vehicle configurations would allow the optimization process to better reflect the variability in human posture. Such refinements could significantly improve the method's ability to predict driving postures across a diverse population.

Conclusion

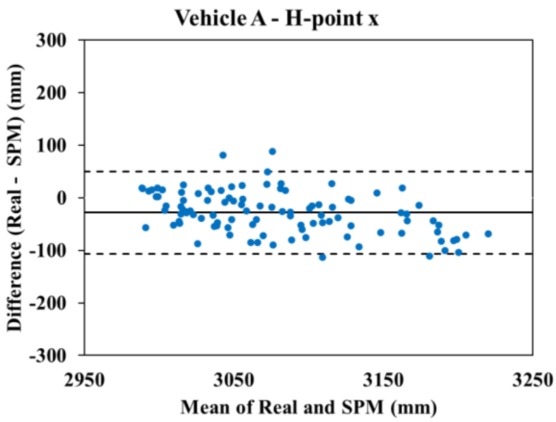
This case study provides an applied comparison of two methods commonly used in the industry for predicting seated postures in occupant packaging design. The SPM and OPM were evaluated using real-world data from two vehicles with contrasting seat configurations. Both methods demonstrate potential viability, but their performance varies depending on the specific tested vehicle configurations and predicted driving position points. SPM, which uses closed-form equations, showed better alignment with measured data for Vehicle B. Its simplicity, ease of use, and computational efficiency make it a practical choice for situations requiring quick predictions without complex simulation setups. In contrast, the evaluation metrics demonstrated that OPM, which uses an optimization approach based on human body models and optimal joint angles, has better alignment for Vehicle A. Yet, the Bland-Alman analysis revealed that SPM, in some cases, has a narrower LOA reflecting more precision despite the systematic bias. OPM has the ability to incorporate a broader range of ergonomic factors, such as joint angles and constraints, providing a comprehensive and flexible approach for occupant packaging design. However, further refinement is required to improve its performance for diverse and specific populations and vehicle configurations.

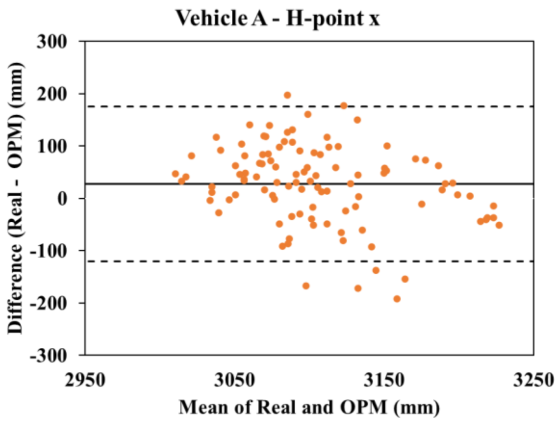
In conclusion, this case study illustrates that both driver posture prediction methods have their strengths and weaknesses, and the specific design context, the vehicle configuration, and the need for precision in certain prediction points should guide the choice between them. SPM is considered ideal for quick predictions and situations where computational simplicity is required, while OPM is considered better suited for more complex occupant packaging designs. Furthermore, these methods should not be limited to drivers but also be extended to the design of seating positions for other occupants in the vehicle, such as passengers. Considering the varying needs and postures of different vehicle occupants is crucial for optimizing comfort and ergonomics across all seating positions. In the context of autonomous vehicles, where driving tasks may no longer remain the primary activity, future research should explore how both SPM and OPM can be adapted for varied occupant positions and activities. Autonomous vehicles offer opportunities for new occupant postures and seating configurations, as users engage in non-driving activities. Thus, incorporating these methods in the design of autonomous vehicle interiors could significantly enhance passenger comfort and safety.

Integrating these methods into the early stages of vehicle occupant packaging design can enhance the overall comfort and safety of vehicle interiors by ensuring that diverse driver populations are better accommodated. Future research should focus on refining both methods, particularly enhancing OPM's adaptability to broader vehicle types, body sizes, and occupant roles (e.g., passengers) and activities (e.g., non-driving tasks), and improving SPM's ability to handle more varied constraints could extend its utility. These refinements will potentially help improve the accuracy of DHM tools and contribute to developing safer and more comfortable vehicle interior designs, particularly as vehicle technologies evolve.

Appendix

This appendix presents Bland-Altman plots for comparing predicted and measured values across key prediction points for SPM and OPM. These plots provide a visual representation of the methods' precision and adaptability, highlighting the differences between predictions and real-world data in the specific design contexts examined in this case study. In each plot, the solid horizontal line represents the mean bias, which is the average difference between the predicted and measured values, indicating whether a method tends to systematically overpredict or underpredict. The dashed horizontal lines represent the limits of agreement (LOA), showing the range within which most differences are expected to fall. Narrow LOA indicate higher precision, while the distribution and pattern of the data points offer insights into each method's adaptability and systematic behaviour.





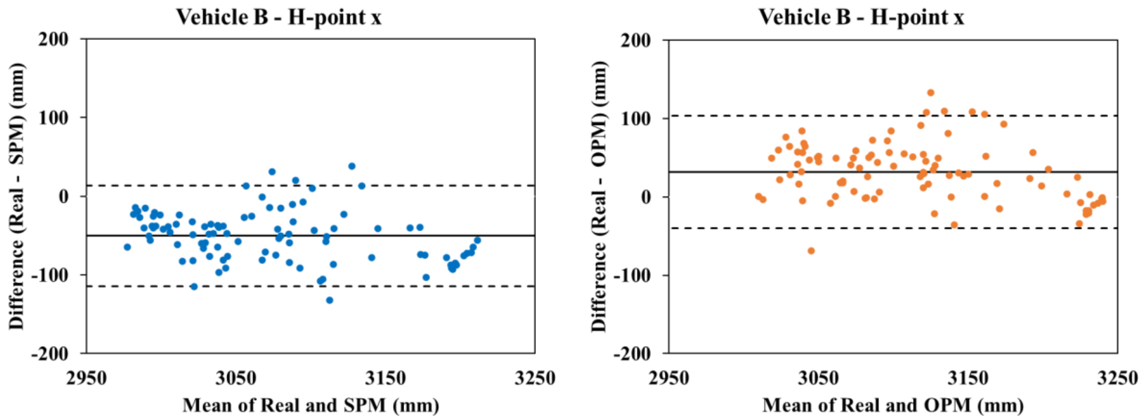


Figure A1. Bland-Altman plots comparing H-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

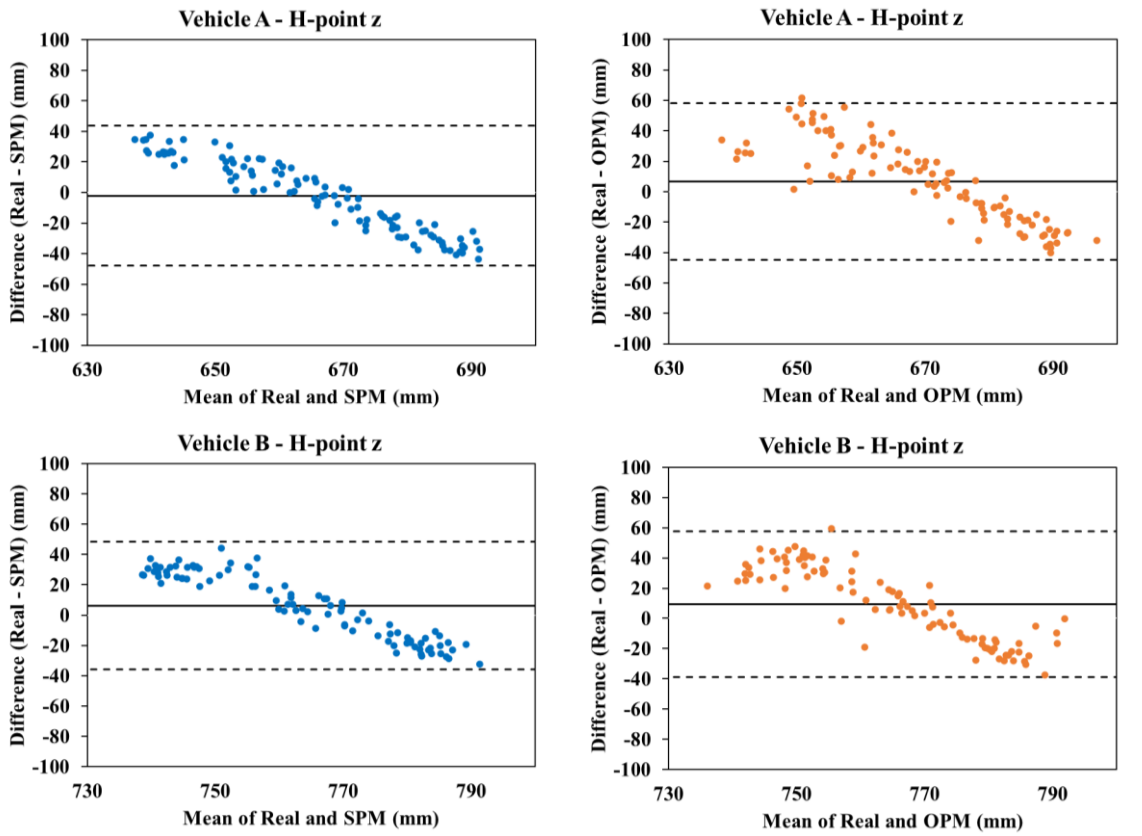


Figure A2. Bland-Altman plots comparing H-point z predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

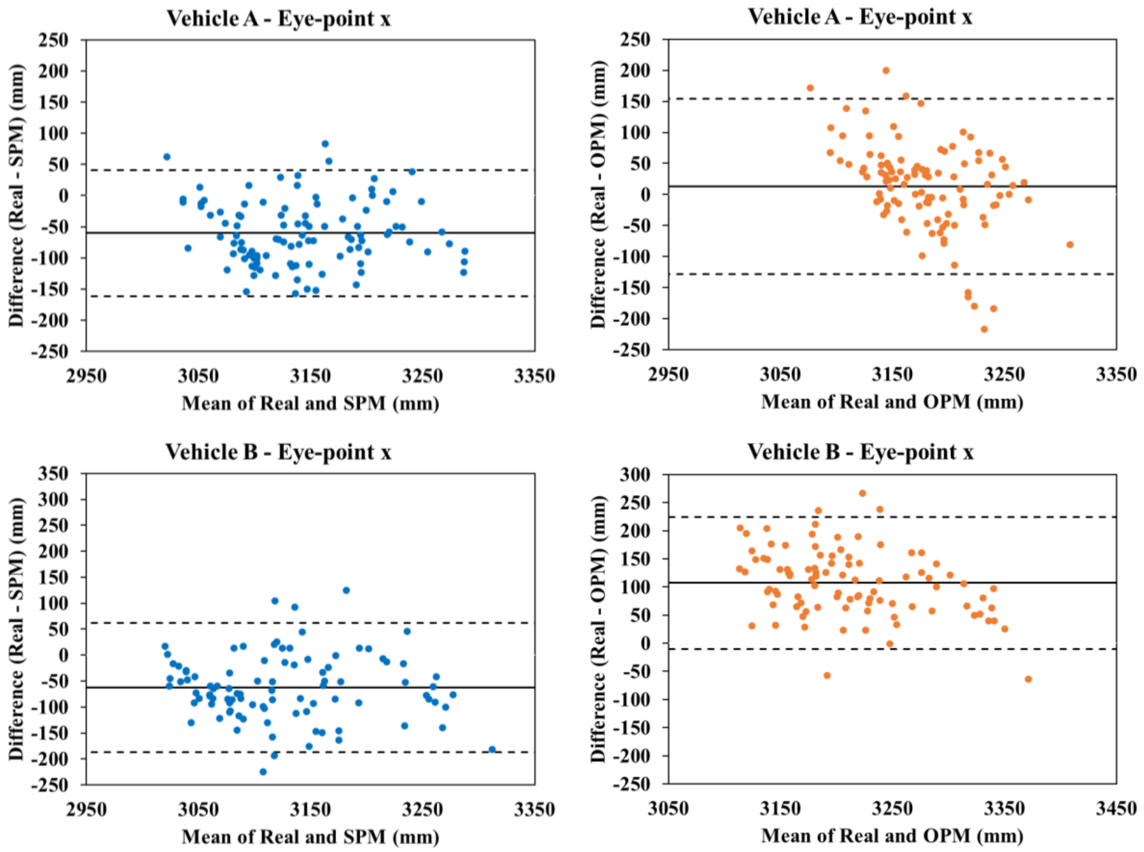
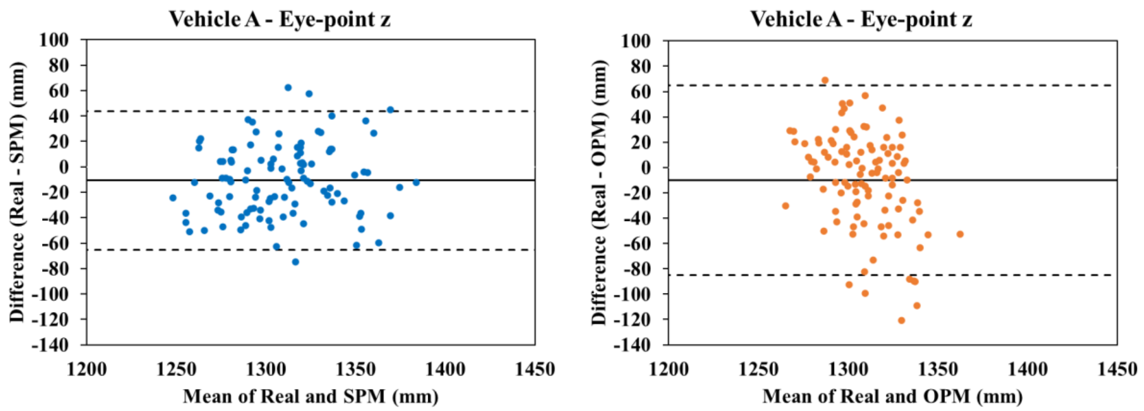


Figure A3. Bland-Altman plots comparing Eye-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).



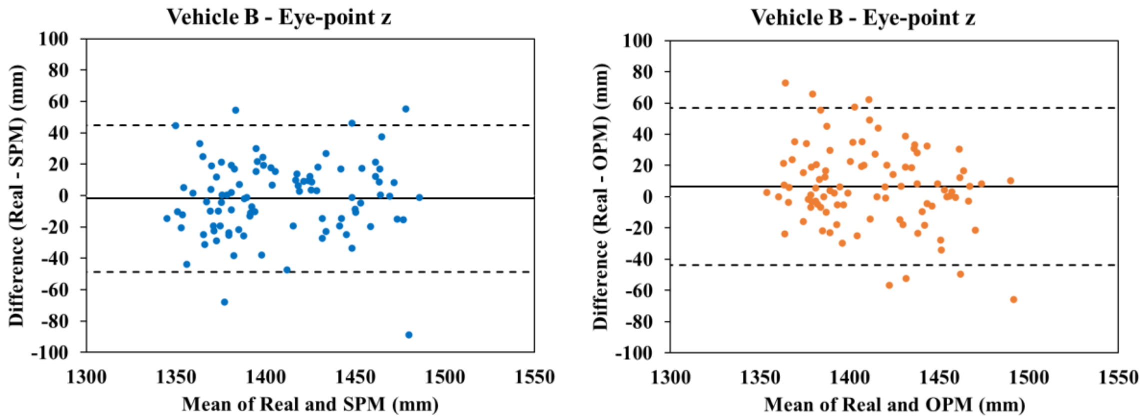


Figure A4. Bland-Altman plots comparing Eye-point x predictions for SPM (left column) and OPM (right column) in Vehicle A (top) and Vehicle B (bottom).

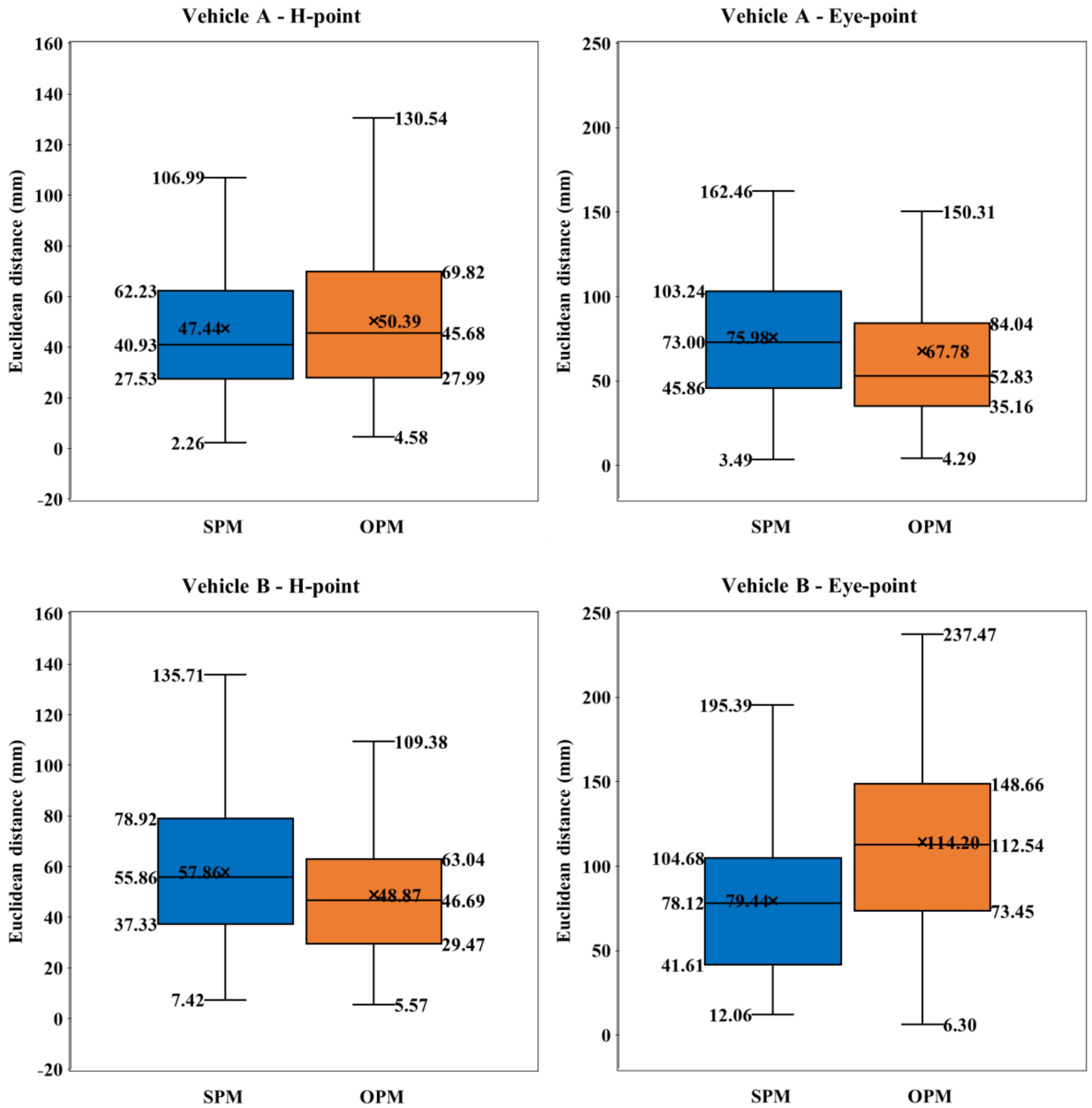


Figure B1. Euclidean distances between predicted and observed values for SPM and OPM across H-point (left) and Eye-point (right) positions in Vehicle A and Vehicle B (bottom).

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